

5	(a)	similarity: e.g. same wavelength/frequency/period, constant phase difference	B1	
		difference: e.g. different amplitude/phase (do not allow a reference to phase for both similarity and difference)	B1	[2]
	(b)	constant phase difference so coherent	B1	[1]
	(c)	(i) $intensity \propto amplitude^2$	C1	
		$I \propto 3^2$ and $I_B \propto 2^2$ leading to	M1	
		$I_B = \frac{4}{9}I$	A0	[2]
		(ii) resultant amplitude = 1.0×10^{-4} cm	C1	
		resultant intensity = $\frac{1}{9}I$	A1	[2]
	(d)	(i) displacement = 0	B1	[1]
		(ii) $x_A = -2.6 \times 10^{-4}$ cm and $x_B = +1.7 \times 10^{-4}$ cm allow $\pm 0.5 \times 10^{-4}$ cm resultant displacement = $(-) 0.9 \times 10^{-4}$ cm	C1 A1	[2]